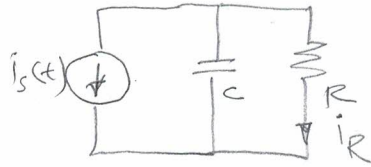


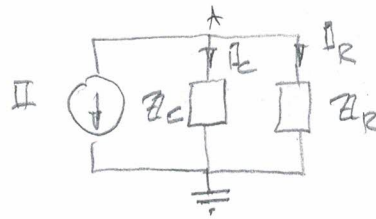
4.59)



$$\begin{cases} i_s(t) = 1 \cdot \cos(200\pi t) \text{ A} \\ C = 100 \text{ nF} \\ R = 100 \text{ } \Omega \end{cases}$$

Bestimmen Strom i_R

Lösung: Heranführen komplexer Representation an Knetzen



Phasor Mod A, KSTL

$$-I - I_C - I_R = 0$$

Nodspannung für V_A

$$\begin{cases} I_C = \frac{V_A}{Z_C} \\ I_R = \frac{V_A}{Z_R} \end{cases} \quad (1)$$

$$\text{Phasor} \quad -I - \frac{V_A}{Z_C} - \frac{V_A}{Z_R} = 0$$

$$\text{oder dann} \quad V_A = - \frac{I}{\frac{1}{Z_C} + \frac{1}{Z_R}} = - \frac{1 \angle 0^\circ}{j200\pi \cdot 100 \cdot 10^{-6} + \frac{1}{100}} =$$

$$= - \frac{100 \angle 0^\circ}{1 + j6.283} = - \frac{100 \angle 0^\circ}{6.36 \angle 80.95^\circ} =$$

$$= -15.72 \angle -80.95^\circ$$

oder Strom i_R via (1)

$$I_R = \frac{V_A}{Z_R} = - \frac{15.72 \angle -80.95^\circ}{100} = -0.157 \angle -80.95^\circ$$

oder Rechenen an Delta

$$i_R = \text{Re}(I_R) = -0.157 \cos(80.95^\circ) \text{ A}$$

$$\underline{\text{Svar: } i_R = -0.157 \cos(80.95^\circ) \text{ A}}$$